

KEY CLAIM

By reading structured neural distortion of individual with Color-Vision deficiency (CVD), we invert it into a personalized color correction filter

BACKGROUND & GAP

- Health phenotypes(cardiac¹, gene-expression²) are often **structured distortions**.
- Such distortions **varies across individuals** even within a diagnostic category.
- Prior work simulates the loss or applies population-level fixes.
 - Inverting a person's own distortion for intervention is under-explored.

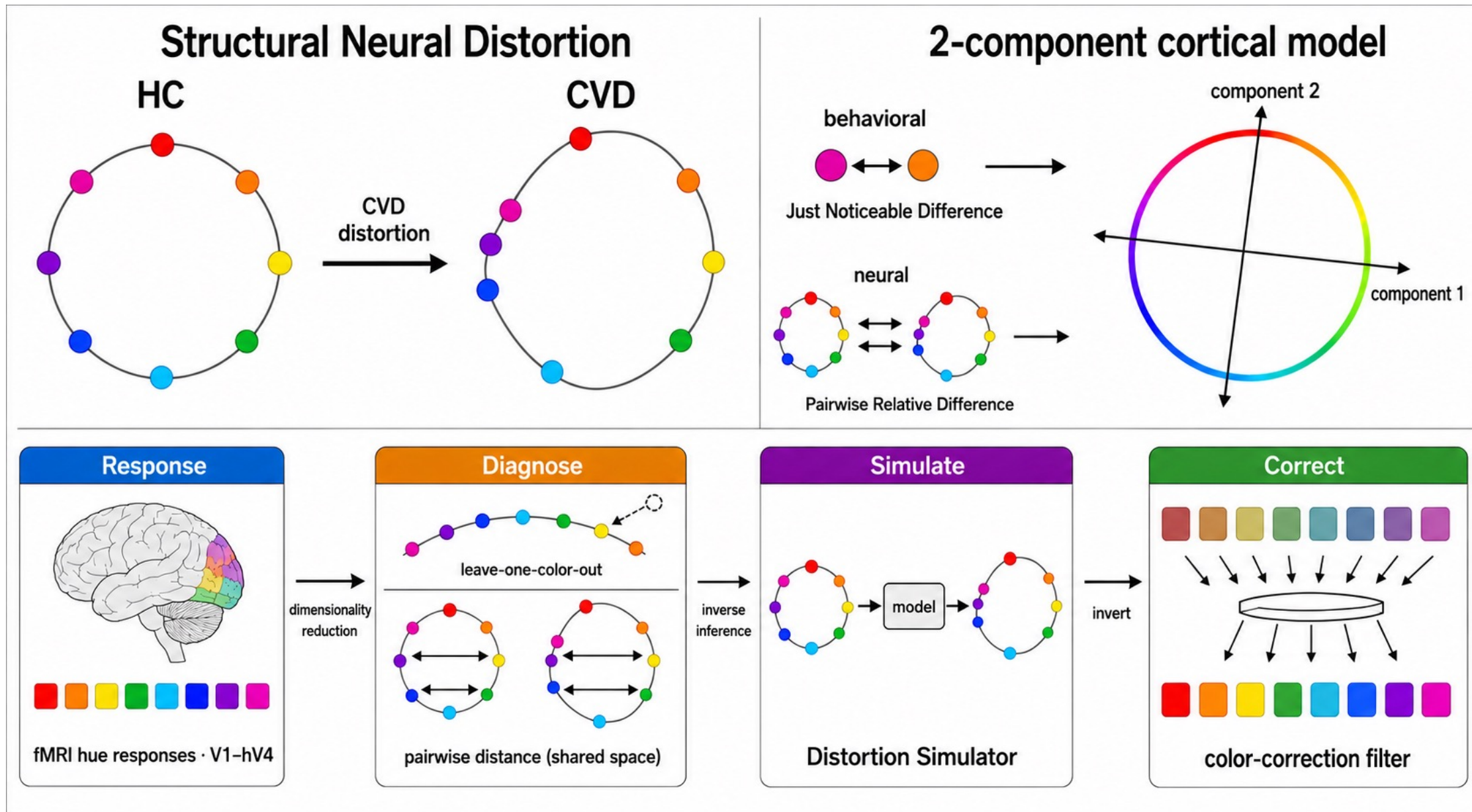
Contributions

- We represent CVD as a distortion of the brain's hue-representation geometry
- We fit a cortical model per person by combining behavioral & neural losses
- We invert each person's fitted model into their color-correction filter.

METHODS – EXPERIMENT & ANALYSIS

- Participants(N = 10)**: 7 Healthy Controls (HC) + 3 CVD (two deutan, one protan)
- fMRI Experiment: 8 isoluminant hues, 6 runs, ROIS V1-hV4**
- LOCO(leave-one-color-out)**: can a held-out hue be rebuilt from the other seven?
 - Forward-Encoding model** maps $\rightarrow$ 6 channels $\rightarrow$ neural responses
 - Tests **continuous hue structure** in neural hue-representation
- RDM(relative distances)**: how far apart are hue pairs in the brain representation?
 - Shared Response model(SRM)** projects HC &CVD into a shared low(4)-dim space
 - Δ RDM = CVD – HC = which pairs are relatively merged or split

FRAMEWORK & MODEL - LOSS



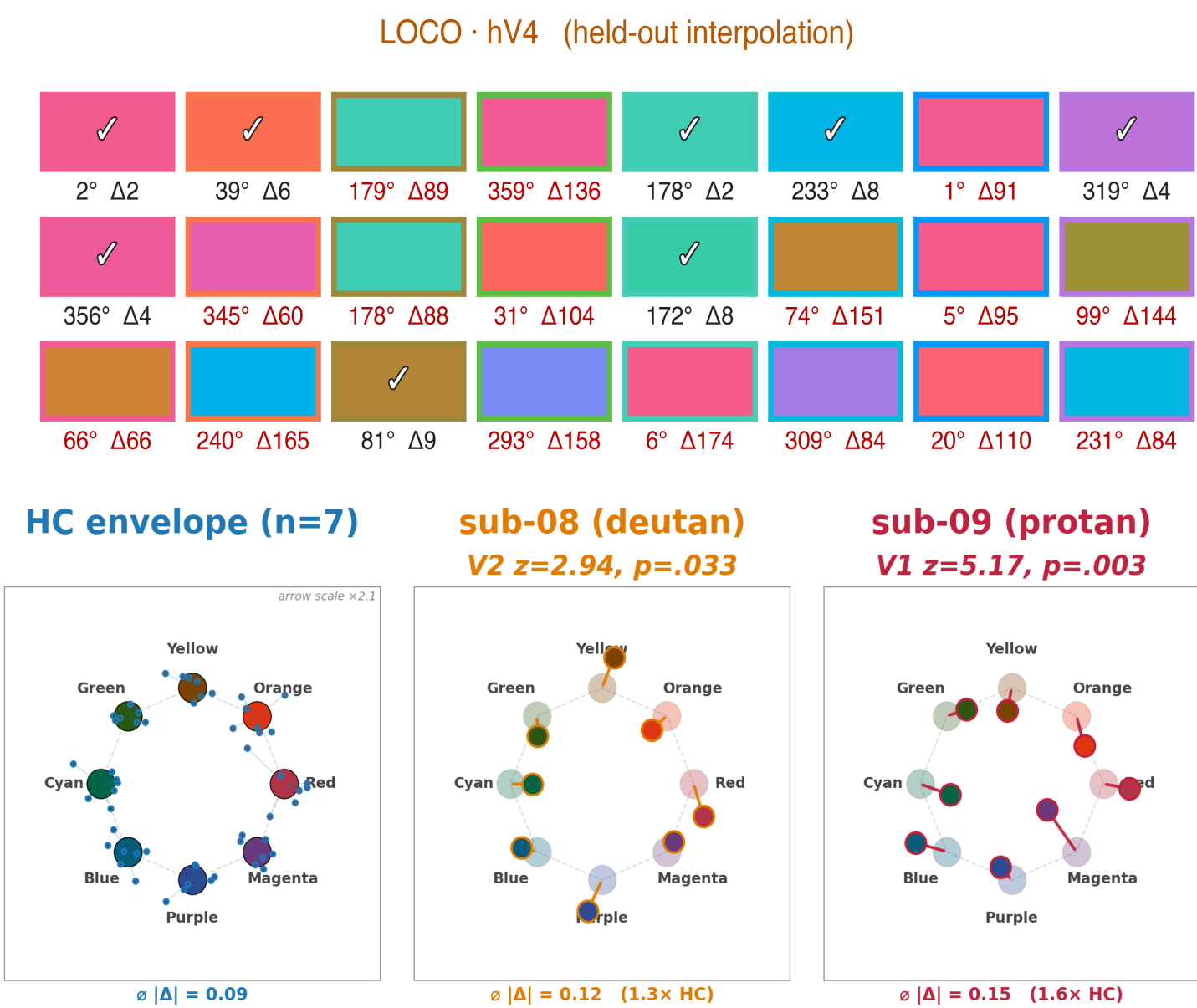
Diagnose each person's hue distortion, fit a cortical model as a deficiency simulator, then invert it into a color-correction filter.

Model mechanism	Parameters	Form
Retinal Cone-Shift ⁶	1	$\Delta\lambda$
Retinal + Cortical compensation	1	$\delta\theta = (2-g)\cdot\delta\theta_Machado$
Cortical two-component	2	$\theta' = \theta + \beta_s \cos(\theta-90^\circ) + \beta_c \cos(\theta-\theta_conf)$

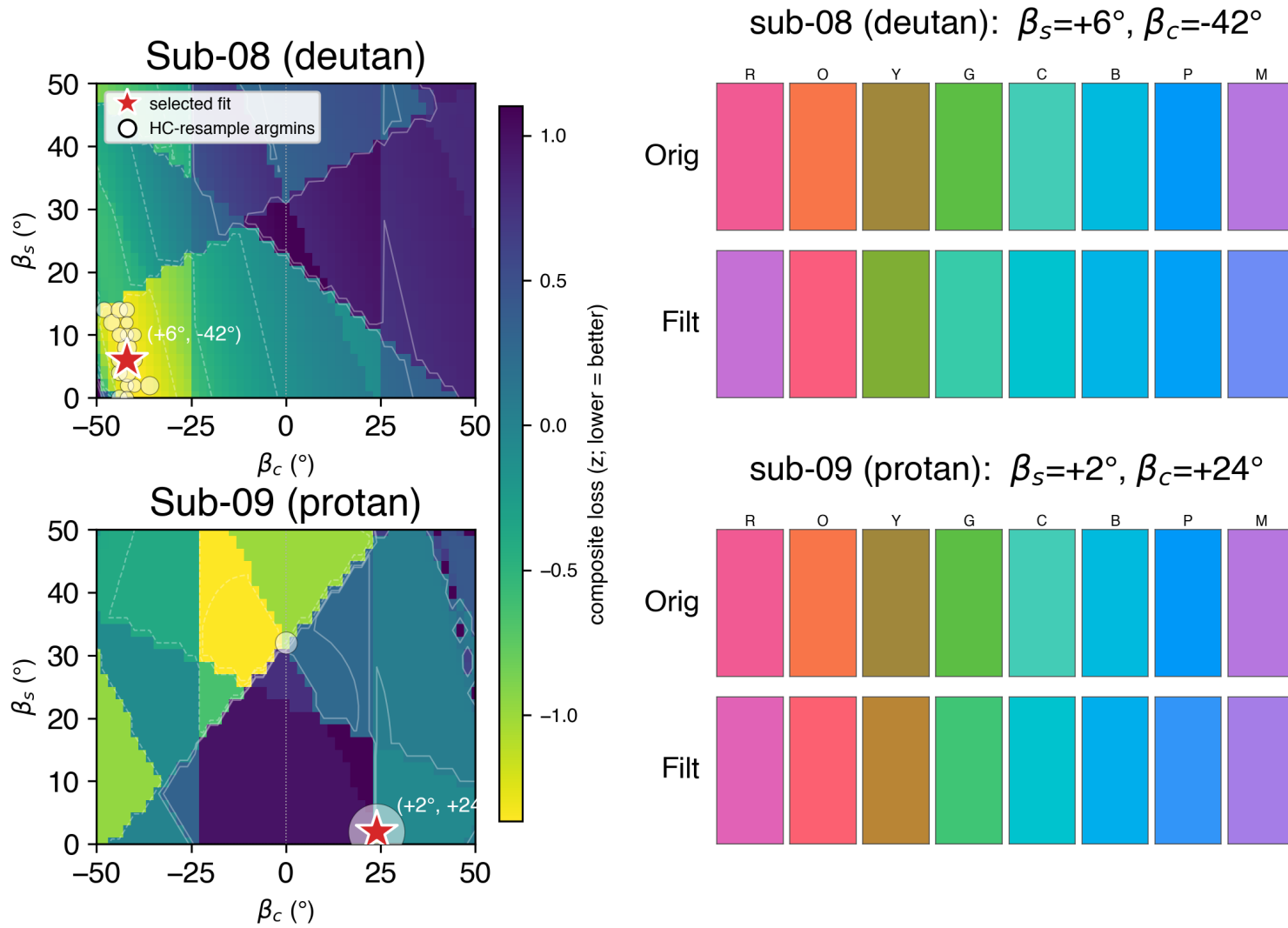
Loss = behavioral JND (Ly) + V1,V2 Δ RDM (L_RDM) + hV4 LOCO (L_LOCO);
selected by held-out generalization.

RESULTS

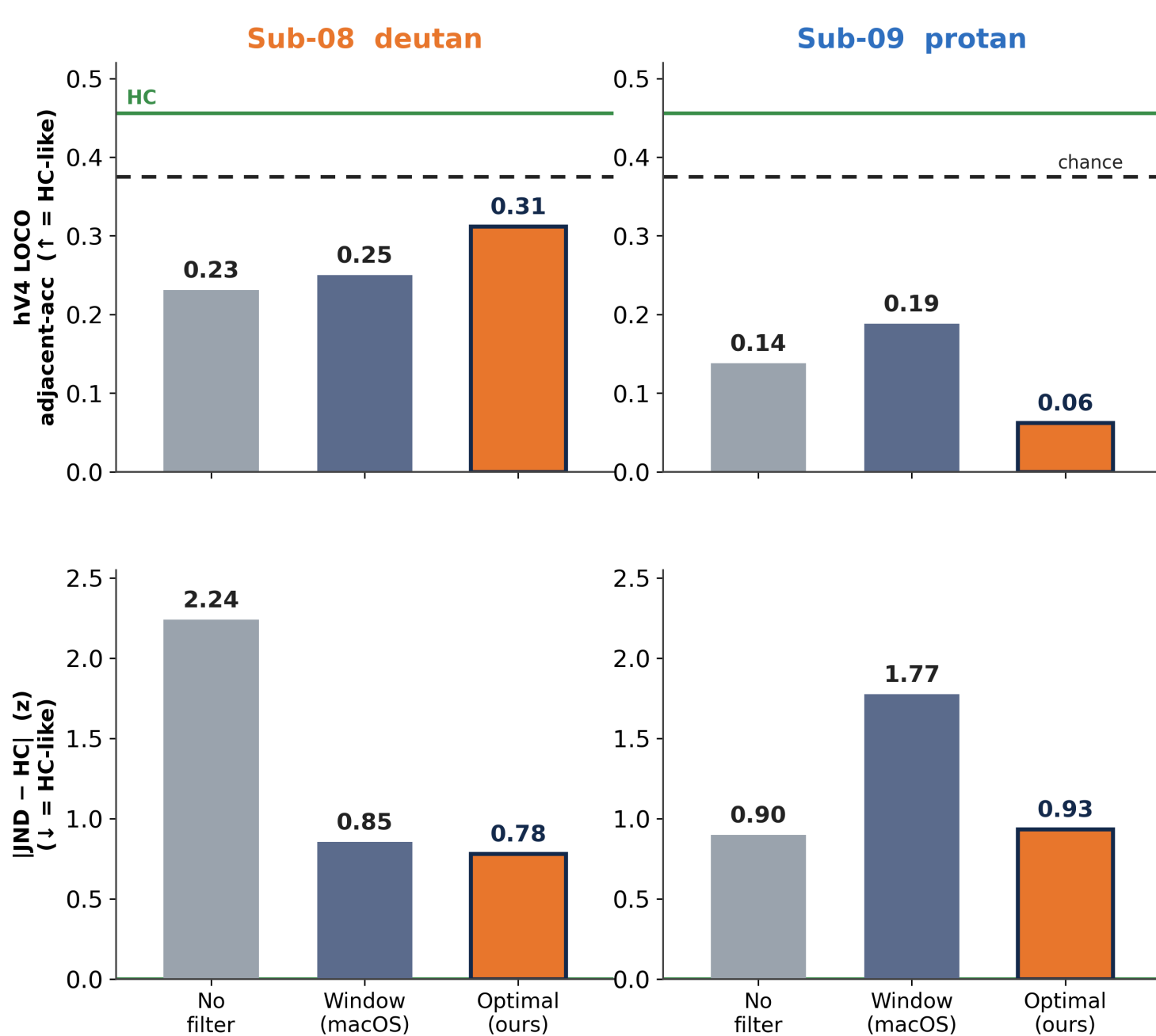
A. Represent · A structured distortion



B. Fit · Deficiency modeling



C. Validation (2nd-MRI, Exploratory)



Category discrimination preserved

- LORO accuracy difference: p=0.668
- Every hue stays decodable.

But continuous interpolation collapses in LOCO

- LOCO Accuracy: HC 0.47
- $\rightarrow$ Deutan 0.25 (p=.082) / Protan 0.13 (p=.024)

The distortion localizes to a subject-specific ROI

- Deutan V2 p=0.040, Protan V1 p=0.007

Retina models failed:

- Overcompensation, non-invertible

Result (β_s , β_c ; final loss):

- Deutan (+6°, -42°); L = $\gamma_{OY} + \Delta$ RDM_V2
- Protan (+2°, +24°); L = $\gamma_{all} + \Delta$ RDM_V1

Neural loss stabilizes parameters

- Deutan: β_c sign stable, no zero-crossing
- Protan: decisive only with neural

(3/7 $\rightarrow$ 7/7)

CONCLUSIONS

End-to-end framework:

- Structured distortion $\rightarrow$ Analytic inversion fit $\rightarrow$ personalized filter

Behavioral HC-likeness favors personalized filter; neural effects exploratory.

- Limits:** N=2; heterogeneous results across metrics/subjects
- Next:** a unified geometry target (Integration across metrics, ROIs), larger N

REFERENCES

- Biffi C et al. IEEE Trans Med Imaging 39, 2088–2099 (2020).
- Lotfollahi M et al. Nat Methods 16, 715–721 (2019).
- Brouwer G, Heeger D. J Neurosci 29, 13992–14003 (2009).
- Kriegeskorte N et al. Front Syst Neurosci 2, 4 (2008).
- Bannert M, Bartels A. J Neurosci 38, 3657–3668 (2018).
- Machado G et al. IEEE Trans Vis Comput Graph 15, 1291–1298 (2009).
- Somers L et al. Vision Res 218, 108390 (2024).
- Tregillus K et al. Curr Biol 31, 936–942 (2021).